Chapter 10
Determining Uncertainties in Fitted Curves



This chapter discusses methods of fitting curves to measured data while accounting for uncertainty in the data. We start with equations for the best linear fit and uncertainties in the slope and intercept. A short discussion of linear fits when there is uncertainty in the x-values is presented next. Confidence and prediction bands are presented for expressing uncertainty in the fitted line and uncertainty in predictions made using the fitted line. Goodness-of-fit metrics are used to assess model adequacy. A summary of techniques for fitting nonlinear equations to data is given, including formulas for nonlinear confidence and prediction bands. In this chapter "nonlinear equation" means an equation that is nonlinear in the independent variable. Finally, a case study applying curve fitting to a calibration drift problem is provided with a short discussion of how to choose optimal calibration intervals based on predicted drift.

10.1 The Purpose of Fitting Curves to Experimental Data

Previous chapters presented the GUM's propagation of uncertainty approach, in which uncertainties of multiple independent variables are propagated through an indirect measurement model $y = f(x_1, x_2, \ldots, x_N)$. In practice it is often useful to take measurements of a dependent variable y over a range of values of an independent variable x. Fitting a line or a curve to the data then provides a model that can be used to predict y, and its uncertainty, for any value of x. In this chapter, the measurement model

$$y_i = f(x_i; p) + \varepsilon_i$$

is used to address this prediction problem. In this expression, y_i is the measured dependent variable and x_i is the independent variable that could be either known (such as time) or measured with uncertainty (such as temperature).

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The term ε_i represents the difference (the residual) between the fitted model and the observed dependent measurement. The goal of curve fitting as discussed in this chapter is to identify the form of the function f and estimate the parameters p that produce the least error (smallest $\varepsilon_i's$) in predicting y given x.

Measurements are often made over a range of physical settings. For example, density may be measured at various temperatures, radiation dose may be measured at various distances from the source, and resistance may be measured at various time intervals to observe drift or decay. All these cases involve characterizing the relationship between two variables. By fitting a curve to experimental data, the relationship between two variables can be described concisely with a single equation. The equation also allows prediction of points on the curve for which experimental data was not measured, that is, interpolating the expected value and uncertainty between measurement points, or in some cases, extrapolating beyond the region of measured values. Expressing the uncertainty in y as a function of x usually results in a more precise (and smaller) estimate of uncertainty. Rather than using a bound on uncertainty in the analysis, the uncertainty is based on the value of x.

Fitting a curve to experimental data generally involves minimizing the distance between a proposed curve and the data. While finding the best-fitting curve can be relatively straightforward, propagating uncertainties through the fitting process can be more complicated.

10.1.1 Resistance vs. Temperature Data

A resistance standard is nominally 100 ohms at a temperature of 20 °C. However, the resistor cannot always be used in a controlled environment and may be subject to a wide temperature range. As such, the resistance was measured at temperatures from 0 °C to 40 °C as shown in Table 10.1 and Fig. 10.1. When the resistor is in use, the temperature is monitored and used to determine the actual resistance value at the time of use. A curve fit may be appropriate in this case, allowing the user to predict the resistance when used at 23 °C, a value not explicitly measured in the table.

Table 10.1 Measured resistance versus temperature

Temperature (°C)	Resistance (Ω)
0	92.648
5	95.675
10	96.923
15	97.732
20	100.473
25	101.258
30	104.426
35	105.484
40	109.237

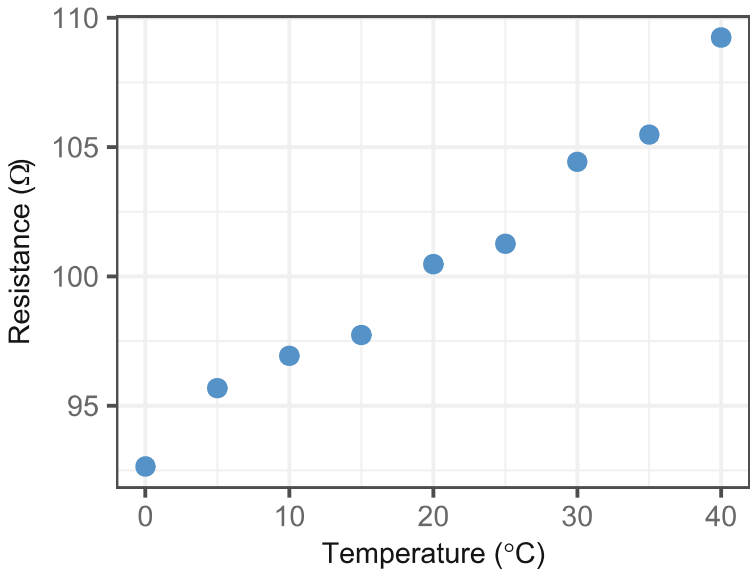


Fig. 10.1 Resistance versus temperature

The sections that follow below use the resistance versus temperature data to illustrate various cases of fitting curves and to describe the uncertainty in those fits.

10.1.2 Considerations When Fitting Models to Data

The first consideration when fitting a model to measured data is determining whether there is even a relationship between the two variables of interest. It is not uncommon to see attempts to fit a line to data that has no real trend. It can sometimes be difficult to distinguish between a relationship between two variables and random noise in the measurement data.

Once a clear relationship is established, the next consideration is choice of a model. In many cases, the simplest possible model is a line. The resistance vs. temperature example has data that appears to fit a line over the range of interest; however, a linear model cannot be extended up or down infinitely—it will eventually break down. For example, when the temperature gets too high, the resistor will melt. From an engineering perspective, a linear model works under reasonable assumptions, but care must be taken when extrapolating the model outside of its valid range. For example, a line fit to the data in Fig. 10.1 may make reasonable predictions in the range from 0 to 40 °C, and a few degrees outside this range, but the same line should not be used to predict resistance at 100 °C. Nonlinear models are often required. For example, an exponential model works well for problems involving decay. Sometimes quadratic or higher-order polynomials work well if the exact physical nature of the model is unknown.

Finally, it should be ensured that there are a sufficient number of data points to define the fitted curve. Two points determine a line exactly, but with no degrees of freedom for error, the uncertainty will be large. Three points determine a quadratic curve that passes exactly through all three points, yet is likely "overfit" resulting in

the curve describing random fluctuations rather than the true physical relationship between the variables.

Several goodness-of-fit metrics are discussed in this chapter that can help quantitatively validate the relationship between variables and proper selection of a model.

In what follows, techniques for finding the best fit parameters (such as slope and intercept) and the uncertainties in these parameters are discussed. Then, formulas for finding the uncertainty in a prediction based on the fitted curve are given, along with some goodness-of-fit metrics. Finally, the cases of nonlinear curve fitting and curve fitting with uncertainties in both x and y values are discussed.

10.2 Methods for Fitting Curves to Experimental Data

The common method for fitting a curve to data is a technique called "least squares fit" or "regression." This method finds the curve that minimizes the vertical distance (technically the sum of the squares of the vertical distances), illustrated in Fig. 10.2, from each measured point to the curve. This can be done analytically for straight line models, but an iterative numerical solution is often required for more complex curves. Most modern computing packages (such as *R*, Matlab, Python/Scipy) have line and curve fitting functions. However, some of the functions do not consider known uncertainty in the measured values or assume an uncertainty based on scatter in the data.

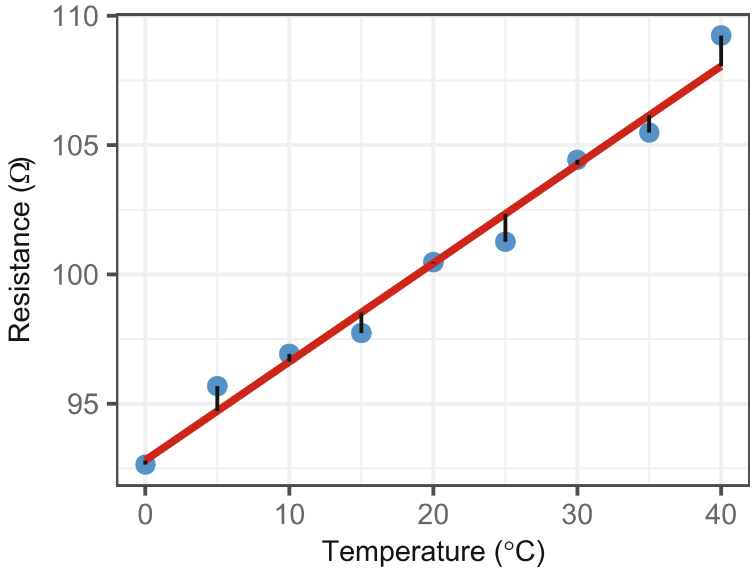


Fig. 10.2 Resistance versus temperature fit line

10.2.1 Linear Least Squares

The linear case of least-squares fitting can be solved analytically. The problem is to find the best fitting line of the form $y = A + Bx$ through the data points. The N measured data points are x_i and y_i.

The least-squares method compares differences between the fit line and each data point to minimize the sum of the squared residuals (the $\varepsilon_i's$), that is, the squared distances from a measured data point to the fitted line. The sum of squared residuals is

$$Q = \sum_{i=1}^{N} (y_i - A - Bx_i)^2. \tag{10.1}$$

This assumes that all the error in the measurement is in the y values. The x values here have no uncertainty. The residuals are minimized by taking the derivative of Q with respect to A and B, setting them equal to zero, and solving the system of equations for A and B. Direct expressions can be obtained:

$$\begin{aligned} A &= \frac{\Sigma x_i^2 \Sigma y_i - \Sigma x_i \Sigma x_i y_i}{\Delta} \\ B &= \frac{N\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{\Delta} \\ \Delta &= N\Sigma x_i^2 - (\Sigma x_i)^2. \end{aligned} \tag{10.2}$$

See Kutner et al. (2005) for a full derivation of these equations. Any uncertainty in the y values is not needed to find the nominal slope and intercept but is used to find the uncertainties of these estimated parameters.

Resistor Example In the resistor vs. temperature data, the best fitting line can be computed using the linear least-squares formulas to obtain a slope of 0.381 Ω/°C and intercept of 92.8 Ω. The line, with the measured data points and residual distances (vertical lines), is shown in Fig. 10.2.

10.2.2 Uncertainty in Fitting Parameters

Uncertainty in the slope B and intercept A can be calculated with Taylor (1997):

$$\sigma_A = \sigma_y \sqrt{\frac{\sum x_i^2}{\Delta}}, \sigma_B = \sigma_y \sqrt{\frac{N}{\Delta}}, \tag{10.3}$$

10.2.3 Weighted Least Squares: Non-constant u(y)

In some practical cases, the uncertainty may change for different measurement points on the line. For example, an ohmmeter could change ranges as resistance increases, where each range has a different known uncertainty value. In this case, every measurement point has its own uncertainty σ_i. The above equations can be modified to weight the least squares appropriately using this value. Assigning weights as $w_i = 1/\sigma_i^2$, the weighted solution becomes (Taylor 1997)

$$\begin{aligned} A &= \frac{\sum w_i x_i^2 \sum w_i y_i - \sum w_i x_i \sum w_i x_i y_i}{\Delta} \\ B &= \frac{\sum w_i \sum w_i x_i y_i - \sum w_i x_i \sum w_i y_i}{\Delta} \\ \Delta &= \Sigma w_i \Sigma w_i x_i^2 - (\Sigma w_i x_i)^2 \end{aligned} \tag{10.6}$$

and uncertainties:

$$\begin{aligned} \sigma_A &= \sqrt{\frac{\sum w_i x_i^2}{\Delta}} \\ \sigma_B &= \sqrt{\frac{\sum w_i}{\Delta}}. \end{aligned} \tag{10.7}$$

10.2.4 Weighted Least Squares: Uncertainty in Both x and y

When the measurements have uncertainty in both x and y, the problem becomes much more complicated. An analytical solution is no longer possible or straightforward, and an iterative numerical technique must be used to minimize the residuals. One good example is the algorithm described by York et al. (2004) and its pseudocode implementation by Wehr and Saleska (2017) which accounts for not only σ_x and σ_y but also the correlation between errors in x and y.

In some cases, it may be simpler to apply a Monte Carlo approach (Robert and Casella 2004) rather than implement a complex algorithm such as this.

Resistor Example Suppose the uncertainty in the recorded temperature is 0.75 °C and the uncertainty in resistance measurements follows 0.5 + 2.5% Ω. These uncertainties are shown as the error bars in Fig. 10.3. Accounting for these uncertainties using the York algorithm, the new slope and intercept values are

$$B = 0.373\ \Omega/\text{cm} \pm 0.025\ \Omega/\text{cm}$$

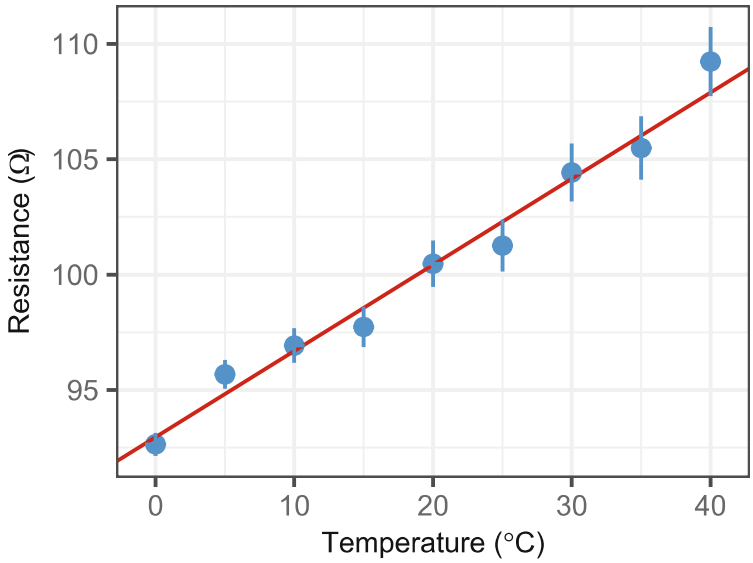


Fig. 10.3 Resistance vs. temperature with uncertainties in both axes

$$A = 92.96\ \Omega \pm 0.43\ \Omega \tag{10.8}$$

```
# Uncertainty in x and y
sdx <- 0.75
sdy <- temperature/40+0.5

# Best-fit straight line
fit <- bfsl(temperature, resistance, sd_x = sdx, sd_y = sdy)
fit
## Best-fit straight line
##
## Estimate Std. Error
## Intercept 92.96163 0.42860
## Slope 0.37299 0.02541
##
## Goodness of fit:
## 0.6418
```

Notice that with σ_x and σ_y accounted for, σ_A is a bit smaller, but σ_B is a bit larger because σ_y is no longer assumed from residuals only.

10.3 Uncertainty of a Regression Line

Two types of uncertainties can be calculated based on fitted lines: the uncertainty in the slope and intercept parameters, and the uncertainty in the line itself.

10.3.1 Uncertainty of Fitting Parameters

The above least-squares methods provide the slope, intercept, and the uncertainty in each, but not the uncertainty in the overall line, i.e., the uncertainty in the predicted value of y for a particular value of x. In the resistor problem, for example, the resistance (y) and uncertainty at a given temperature (x) cannot be obtained directly from the slope and intercept and their uncertainties.

10.3.2 Confidence Bands

Just as the mean of a measurement sample is an estimate of a variable's true mean, the fitted line $y = A + Bx$ is an estimate of the true line relating the two variables. Analogous to the standard error of the mean, the confidence band is a measure of error in the fit line. Because it encompasses all the possible lines that might fit the data, and the line is best defined near the center of the data points, the confidence band is a function of x and is typically smallest near the center of the line and largest on the ends. The confidence band as a function of x is

$$u_{\text{conf}}(x) = \sigma_y \sqrt{\frac{1}{N} + (x - \bar{x})^2 \left(\frac{\sigma_B}{\sigma_y}\right)^2}, \tag{10.9}$$

and the true line, with k-factor applied, is expected to fall in the range $f(x) \pm k \cdot u_{\text{conf}}(x)$ with high confidence.

10.3.3 Prediction Bands

A prediction band is used to determine the uncertainty in a prediction associated with an individual measurement (x) on the fitted curve. This band includes both uncertainty in the fitted line (i.e., the confidence band) and uncertainty in the individual measurement. The prediction band can be calculated from the RSS of the confidence band with the σ_y uncertainty estimated from the residuals:

$$u_{\text{pred}}(x) = \sqrt{\sigma_y^2 + u_{\text{conf}}^2(x)} \tag{10.10}$$

Doing a little algebra:

10.4 How Good Is the Model?

A simple line may fit the data reasonably well and still not be the best model for the data. Other models may fit the data better or make more physical sense. Some qualitative and quantitative tests can be used to determine if the most appropriate model was chosen. Those tests are discussed in this section.

10.4.1 Residual Analysis

The residual plot (plotting the $\varepsilon_i's$ vs the $x_i's$) is one of the key diagnostic tools for assessing a fitted model. For any reasonable fit of the data, the individual measurements should fall randomly above and below the curve. The residuals (distances from each point to the fitted curve) should also be roughly normally distributed with mean zero. This assumption can be tested via a simple histogram or normal probability plot. A good linear fit and residual plot is shown in Fig. 10.5a.

Now consider Fig. 10.5b. The data show a slight curvature when compared to the linear fit. The curvature is not always as obvious as in this example, so a better qualitative test is checking the residual plot. Note the order of the residuals, with positive values occurring near the ends and negative values in the center. This indicates a straight line model may not be the best fit for this data. Perhaps a polynomial or exponential curve should be considered.

The data in Fig. 10.5c show an example with a single large outlier. The large outlier greatly influences the slope of the fitted line, and the majority of the data are not well fit by the line. Again, using a residual plot shows clearly that the data are not well fit, and the residuals are not randomly distributed about zero.

10.4.2 Slope Test

In fitting simple lines, another good check of the fit is the slope test. This test can be used to confirm a linear relationship between the x and y variables. If there is indeed a linear relationship, the slope of the line will have a statistically significant positive or negative slope. To apply the slope test, check the 95% coverage interval of the slope and its uncertainty (Eq. 10.5). If this interval contains zero, the test fails, and one cannot conclude there is a linear relationship between the x and y variables (Fuller 1987). The slope test is equivalent to testing that the (x, y) correlation coefficient is statistically different from zero (Glantz and Slinker 2001). The validity of these tests relies on the assumptions regarding the residuals as stated in Sect. 10.4.1.

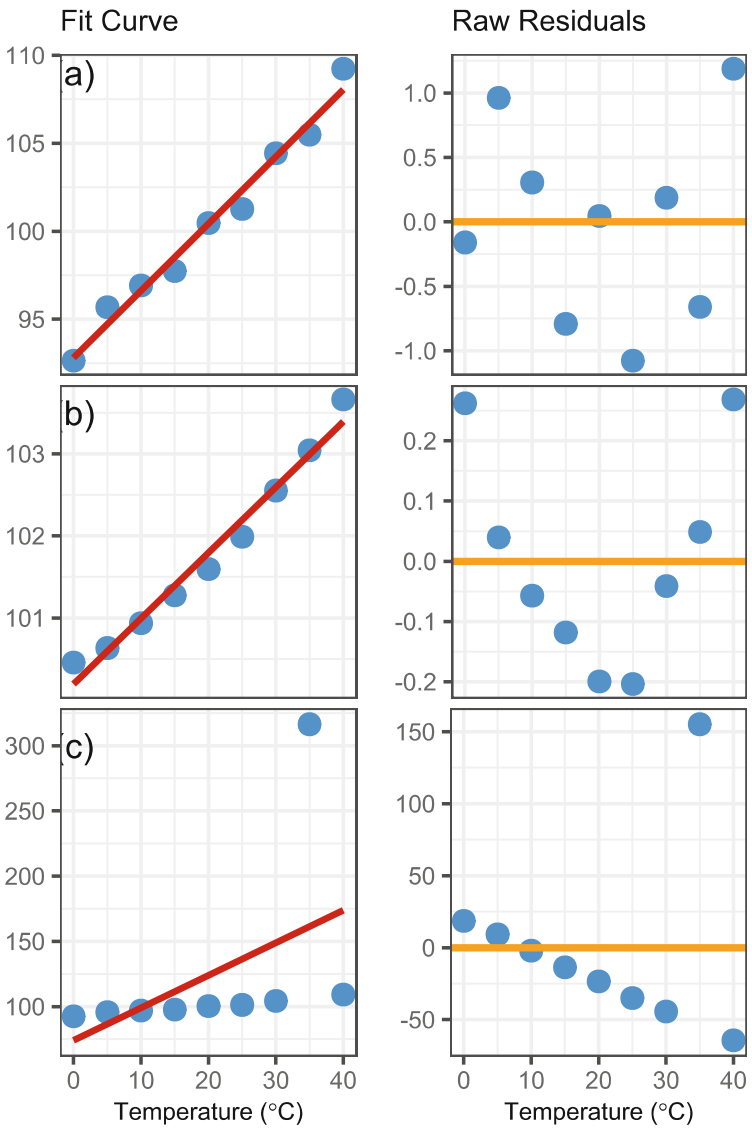


Fig. 10.5 Residual analysis examples

Similar tests can be used to check the validity of certain nonlinear models. For example, a test of the parameter A in the model $y = \exp(Ax)$ can be used to confirm an exponential relationship between the x and y variables. On the log scale, the parameter A becomes a slope parameter, and the slope test applies.

10.4.3 Quantitative Residual Analysis

In some cases, a more quantitative approach is desired to assess the goodness-of-fit. Many statistical tests for goodness-of-fit are described by Glantz and Slinker (2001), but one of the most common is the coefficient of determination, or r^2 value. We define r^2 by

Under the assumption that the residuals are normally distributed with constant variance, approximately 95% of the measured values (the $y_i's$) should fall within $\pm 2S$ of the fitted line (the corresponding $\widehat{y}_i's$). The value of S can be used to compare competing candidate models, with models having lower values of S generally being preferred.

In the resistance vs. temperature example from Table 10.1, the r^2 value is 0.97, indicating a very good linear fit. The value of S is 0.82 Ω, meaning that 95% of the measurements are expected to fall within 1.64 Ω of the fitted line.

10.5 Uncertainty in Nonlinear Regression

The previous sections examined fitting lines (slope and intercept) to measurement data. This section generalizes the method above to fitting nonlinear curves. In this chapter "nonlinear" means nonlinear in the independent variable x.

10.5.1 Nonlinear Least Squares

The same least-squares methods can be adapted to nonlinear curve fitting. Recall that the sum of squared residuals (Eq. 10.1) in the linear case has the form:

$$Q = \sum_{i=1}^{N} (y_i - A - Bx_i)^2. \tag{10.20}$$

In the nonlinear case, the distance between the measurements and fitted curve is still used, but the minimization problem takes the more general form:

$$Q = \sum_{i=1}^{N} [y_i - f(x_i; p)]^2, \tag{10.21}$$

where f is the nonlinear function relating x to y, and p represents a vector of parameters that are estimated to minimize Q. Typically, an iterative numerical solution is required to minimize Q using an algorithm such as gradient descent, Gauss–Newton Method, or the Levenberg–Marquardt Method (Gavin 2017).

Resistor Example It was decided to calibrate the resistance standard over a wider temperature range. After taking new measurements up to 100 °C, a new line was fit, shown in Fig. 10.6; however, the residual analysis indicated problems with the linear model.

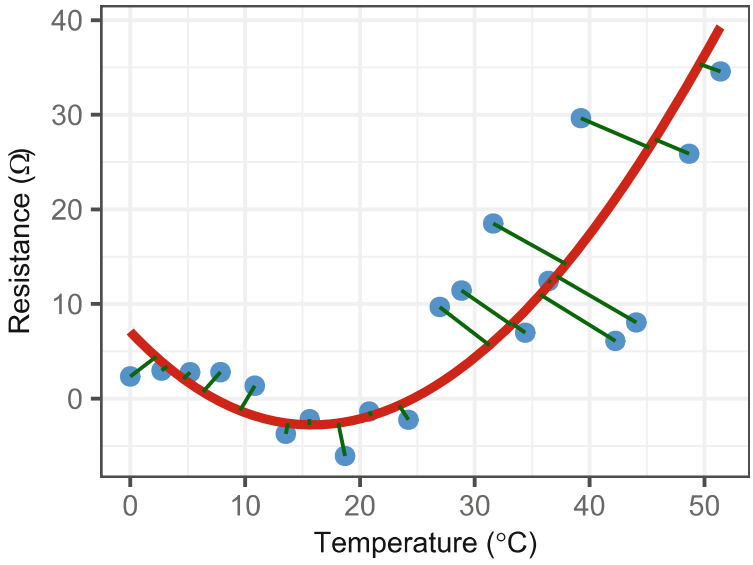


Fig. 10.8 Orthogonal distance regression

```
# Generate data
T2 <- seq(0, 95, by=5)
R2 <- 100 + (T2)**2 / 10000 - .04
R2 <- rnorm(length(T2), R2, 0.04)
# Fit quadratic model
fit <- lm(R2 ~ poly(T2,2, raw = T))
summary(fit)$coefficients
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.000112e+02 2.298894e-02 4350.404352 7.658464e-53
## poly(T2, 2, raw = T)
1 -2.538013e-03 1.121652e-03 -2.262745 3.703469e-02
## poly(T2, 2, raw = T)
2 1.250023e-04 1.139747e-05 10.967549 3.933882e-09
```

10.5.2 Orthogonal Distance Regression

An alternative to least-squares regression, useful when fitting a nonlinear curve with uncertainty in both x and y, is Orthogonal Distance Regression (ODR). It takes the same approach as least-squares regression but instead of minimizing the sum of vertical distances between each point and the curve, it minimizes the sum of shortest orthogonal distances (Fig. 10.8).

```
# Generate data
xo <- seq(0, 50, length.out = 20)
yo <- .0005 * xo^3 -.01 * xo^2 - .05 * xo + 3
uxo <- seq(.1,1, length.out = length(xo))
uyo <- seq(1,10, length.out = length(yo))
yo <- yo + rnorm(length(yo), 0, uyo)
```

```
xo <- xo + rnorm(length(xo), 0, uxo)
dat <- data.frame(xo, yo)
# Fit orthogonal nonlinear least-squares regression model
fit <- onls(yo ~ b1 + b2*xo + b3*xo^2 + b4*xo^3, data = dat, start = list
(b1=.05, b2=-.05, b3=.2, b4=3))
summary(fit)$coefficients
## Estimate Std. Error t value Pr(>|t|)
## b1 2.7102953151 4.3367491296 0.6249601 0.5408084
## b2 -0.1242620821 0.7614549688 -0.1631903 0.8724120
## b3 -0.0105391979 0.0355942465 -0.2960927 0.7709678
## b4 0.0005111474 0.0004619429 1.1065162 0.2848629
```

Unlike ordinary least-squares regression, ODR produces the same curve regardless of which axis is considered the independent variable. In other words, using ODR to fit a line to data (x_i, y_i), regressing x on y, will generate an equivalent result as fitting a line to (y_i, x_i), regressing y on x. Ordinary least squares in the first case would minimize residuals in the y direction and in the second case minimize residuals in the x direction, leading to different results. This is illustrated in Fig. 10.9.

For details of the algorithm, see Boggs and Rogers (1990). Implementations of ODR exist in computing packages such as *R*, Matlab, and Python/Scipy.

10.5.3 *Confidence and Prediction Bands in Nonlinear Regression*

The confidence and prediction bands given above were derived for the special case of linear regression. In the nonlinear case, the confidence interval may be found by approximating the function using a Taylor series expansion and using the delta method as described in Cox and Ma (1995):

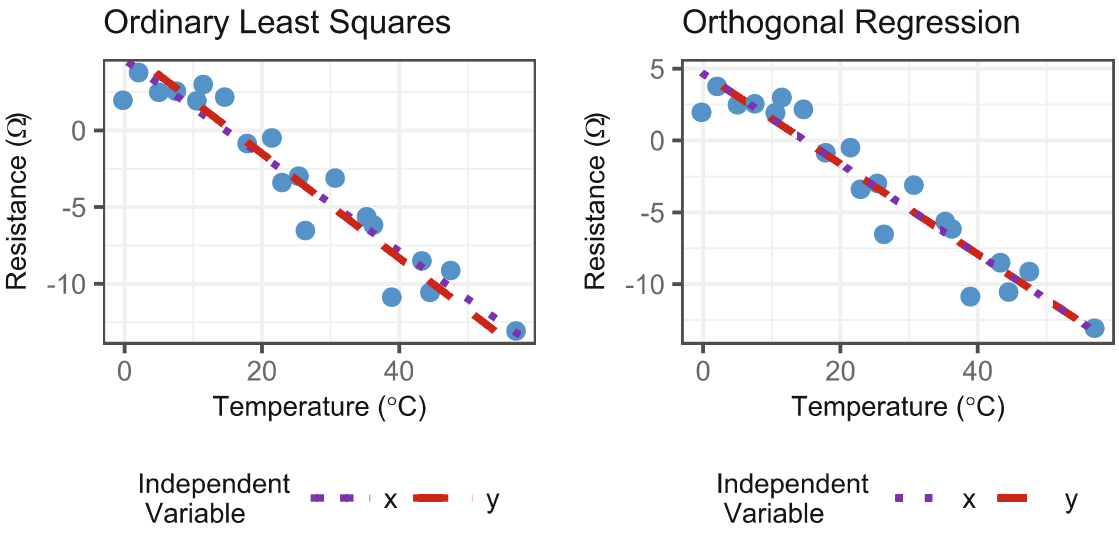


Fig. 10.9 Independent variable in ODR compared to ordinary least-squares regression

$$u_{\text{conf}}(x) = \sqrt{\nabla f^T \cdot C \cdot \nabla f}, \tag{10.23}$$

where ∇f is the gradient matrix of the function at a particular x and C the covariance matrix of the input variables.

Prediction can then be found as before by combining the confidence band with the uncertainty in an individual measurement:

$$u_{\text{pred}}(x) = \sqrt{u_{\text{conf}}^2 + \sigma_y^2}. \tag{10.24}$$

Note that σ_y may also be a function of x and may not be explicitly known for all possible x values.

10.6 Using Monte Carlo for Evaluating Uncertainties in Curve Fitting

Sometimes it is easier to apply a simple least-squares approach to find the best fit parameters, and then use a Monte Carlo method to propagate the uncertainties through the fitted curve. This becomes especially true when a programming language or computing package has a least-squares method that does not allow the input of uncertainties into the independent variables, or only allows a fixed uncertainty to apply to all measurement values. Monte Carlo can also be used to relax the assumption that each measurement value is normally distributed.

10.6.1 Monte Carlo Approach

In the Monte Carlo approach to estimating uncertainty in a fitted curve, each data point (x_i) is considered an input to the model. For each Monte Carlo sample, each data point is independently sampled from its distribution. The distributions for each data point do not have to be the same and can be drawn from any probability distribution. Perhaps there is greater uncertainty when measuring low values of resistance than measuring high values of resistance, for example. Then, with this set of sampled data points, one possible fitted curve is found using the least-squares fit (with no uncertainty input). The sampling process is repeated N times to generate N possible fitted curves and the parameters are estimated for each curve. This results in a Monte Carlo distribution and summary statistics for each parameter (e.g., the mean and standard deviation of possible slopes and intercepts) (Fig. 10.10).

Using Monte Carlo methods for curve fitting reduces the complexity of the calculation. The curve-fitting algorithm does not need to include uncertainty as an input and does not need to produce uncertainties as outputs. If developing code

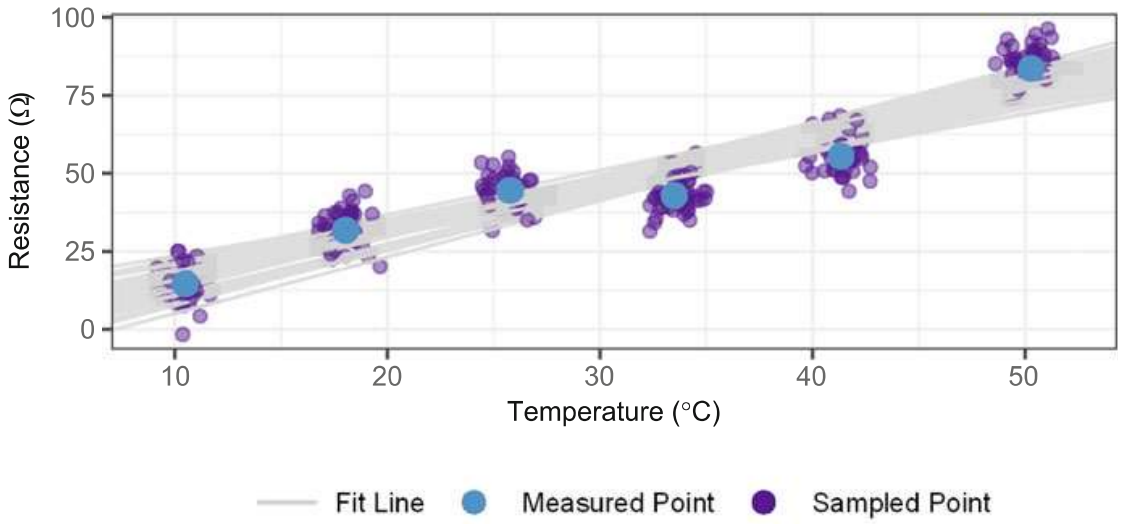


Fig. 10.10 Monte Carlo for curve fitting

from scratch, a basic least-squares fit with no uncertainty consideration can be much easier to implement. The Monte Carlo method does not require the uncertainties in data points to be normally distributed or even identically distributed. Little additional effort is required to account for uncertainty in both the x and y values of each point, a factor that makes analytical methods very complex.

An obvious disadvantage is computation time. A least-squares method on a nonlinear model often requires an iterative (slow) solution, and with Monte Carlo, the slow solution must be computed many times to converge on a representative set of fit parameters. However, depending on the number of data points, "slow" may mean computing the solution in tens of seconds rather than milliseconds, still perfectly acceptable for offline data analysis.

10.6.2 Markov-Chain Monte Carlo Approach

In certain curve fitting problems where the model is highly nonlinear or the variance is unknown, a Bayesian approach called "Markov-Chain Monte Carlo" (MCMC) may be more suitable. With this approach the data points ($x_i's$) remain fixed but the actual fitting parameters (such as slope and intercept for a fitted line) are varied during each Monte Carlo sample (Brooks et al. 2011). A slope and intercept value are randomly generated from an expected distribution and the probability that those parameters produce the best-fitting curve is calculated. New samples of slope and intercept are generated based on probabilities, generating a "random-walk" through the expected parameter space. After N trials, a distribution of slope and intercept parameters can be constructed and summary statistics for each parameter can be evaluated.

10.7 Case Study: Contact Resistance

A common way of measuring the resistance between a metal contact and semiconductor starts by making a series of identical contacts spaced at different lengths along the semiconductor. By measuring the total resistance between each pair of contacts and extrapolating back to zero length, the resistance due to only a single pair of contacts can be estimated (Fig. 10.11).

Table 10.2 lists resistance measurements made between various Titanium-Gold contacts at different lengths along a silicon nanowire. The goal is to find the intercept of the best-fit line.

As displayed in Fig. 10.12, extrapolating back to zero length should provide the combined resistance of two contacts. It is straightforward to calculate A and B for the least-squares line by hand using Eq. (10.2).

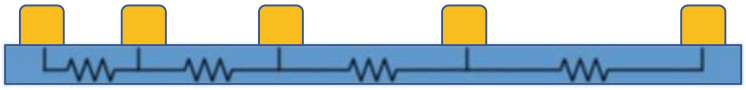


Fig. 10.11 Schematic for measuring contact resistance

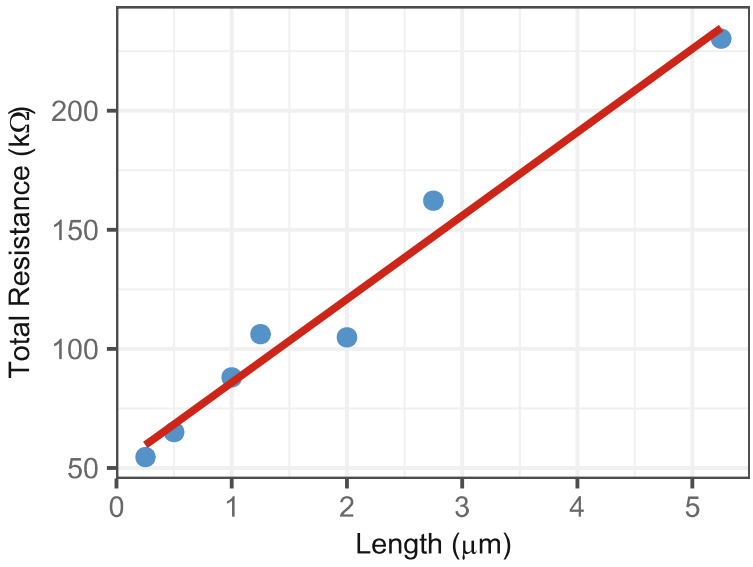


Fig. 10.12 Nanowire resistance vs. length for contact resistance determination

Table 10.2 Contact resistance data

Length (μm)	Resistance (kΩ)
0.25	54.6
0.50	65.1
1.00	88.1
1.25	106.2
2.00	104.9
2.75	162.3
5.25	230.3

measurement, the voltage accuracy is 0.018% + 150 μV and the current accuracy is 0.05% + 100 pA. Using the GUM method described in Chap. 6 for the measurement model $R = V/I$ with these uncertainty values, the total resistance uncertainty is 470 Ω.

This value can be used in place of the σ_y calculated above from the residuals. Now, solving for σ_A gives 270 Ω. This value is much smaller than the 6.7 kΩ calculated before. In this case, the residuals predict a much higher uncertainty than simply accounting for the published uncertainty of the measurement instrument. It is likely there are other uncertainties to consider here in addition to the 470 Ω of the measurement instrument.

Suppose a nanowire device is to be constructed with 4 μm separation between contacts. What is the expected total resistance? The solution involves predicting a value for a new measurement from the fit line; thus, the prediction band must be used. Calculating u_{pred} with $\sigma_B = 2.76$ kΩ, $S_{xy} = \sigma_y = 11.66$ kΩ, and $N = 7$:

$$u_{\text{pred}} = 13.79 \text{ k}\Omega, \tag{10.29}$$

and the total resistance of a 4 μm length of semiconductor with two contacts, to 95.4% confidence ($k = 2$), is estimated to be

$$R_c = 190 \text{ k}\Omega \pm 28 \text{ k}\Omega. \tag{10.30}$$

10.8 Drift and Predicting Future Values

There are additional issues to consider when calibrating devices that drift or decay predictably over time. The same line or curve fitting techniques discussed in this section apply, with the x-axis being time. In typical applications, the fitted curve is extrapolated into the future to predict a value and uncertainty during the time of use.

Consider a resistor on which 15 measurements have been made over the course of 15 years and in which the resistance value appears to be drifting linearly. As always, caution must be used when choosing a model. The resistor cannot have downward linear drift forever or the resistance would eventually become negative! However, in a reasonable lifespan of the device, the linear model likely is sufficient.

Using the methods previously described, a fitted line and its prediction band can be computed for the drifting resistance (Fig. 10.13).

10.8.1 Uncertainty During Use

The resistance value and uncertainty for the latest measurement (Year 14) are presently known, but the resistor will be used over the course of one more year

$$\sigma_c^2 = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} u_{\text{conf}}^2(t) \mathrm{d}t \tag{10.36}$$

and mean uncertainty in $y(t)$ over the interval:

$$\sigma_{\widehat{y}}^2 = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \sigma_y^2(t) \mathrm{d}t, \tag{10.37}$$

leading to a total reported uncertainty, valid for the full calibration interval, of

$$u_R = \sqrt{\sigma_{\bar{b}}^2 + \sigma_c^2 + \sigma_{\widehat{y}}^2}. \tag{10.38}$$

Using this method, the resistance value is reported at the average time during the interval:

$$R = R\left(t_{\text{avg}}\right). \tag{10.39}$$

Often the rate has units of drift per year, and the interval is 1 year ($t_1 = 0, t_2 = 1$), which simplifies this expression further. (For example, $\sigma_{\widehat{y}}^2 = \sigma_y^2$ and $\sigma_{\bar{b}}^2 = r^2/12$).

This method produces a result not as accurate as reporting the values as functions of time (Method 1), but not as conservative as using worst-case estimates (Method 2). It will produce an average OOT rate of 5% when u_R is expanded to 95% coverage under normal rate and uncertainty conditions (Delker et al. 2020).

When using any of the above methods to account for drift, the method and assumptions must be explicitly stated.

10.8.2 Validating Drift Uncertainty

Every time a new calibration is made, the fitted line and uncertainty bands must be recomputed to include the new data point. The model itself should be evaluated after each new measurement to ensure a linear fit is still adequate. After many calibration cycles, the prediction band can become quite large due to the $(t - \bar{t})$ term increasing.

Exercise caution when applying linear drift to devices that may not be predictably drifting. Data that may look like drift could also be random fluctuations due to other uncertainties. Use the slope test to validate whether the drift rate (slope) is significantly different than zero. Attempting to fit a drift line is especially problematic when a device has only been calibrated a few times. Generally, there is much larger uncertainty in the slope of a line going through 3 data points than a line going through 50, due to less spread in the t values and smaller degrees of freedom for estimating the expanded uncertainty. As a general rule, a minimum of 5 measurements are necessary for the slope test to reliably indicate whether the device is drifting.

10.8.2.1 Type B Uncertainty

The Type B uncertainty comes from the resistance bridge used for the measurement, a 1-ohm reference standard calibrated by NIST, and fluctuation in environmental conditions. Other uncertainties, such as uncertainty due to leakage current, were deemed insignificant compared to the other components and were omitted from this analysis. The Type B uncertainty components are listed in Table 10.3.

All components in this table have been normalized and converted to $k = 1$ standard uncertainties. For example, the bridge uncertainties as reported by the instrument specifications were interpreted as uniform distributions and were standardized by dividing by the square root of 3.

10.8.2.2 Type A Measurement Uncertainty

After waiting 30 days for a resistor under test to stabilize in the oil bath under the laboratory environmental conditions, multiple repeated measurements were made over the course of 5 days spread over a 2-week period. Each day's measurement consisted of 40 individual measurements averaged together.

The combined standard deviation is found by pooling the standard deviations of the 5 times 40 individual measurements, grouped by day:

Table 10.3 Type B uncertainty components

Uncertainty	Value ($k = 1$) μΩ/Ω	Description
Reference standard resistor	0.01	Time-of-test uncertainty as calibrated by NIST
Reference standard drift allowance	0.006	Determined using same drift allowance method as described below, but on the reference standard
Resistance bridge uncertainty	0.013	1 Ω specifications for MI-6010C bridge
Resistance bridge linearity	0.0029	1 Ω specifications for MI-6010C bridge
Resistance bridge resolution	0.00029	1 Ω specifications for MI-6010C bridge
Temperature fluctuation	0.0021	Includes oil bath stability and temperature coefficient of standards
Pressure fluctuation	0.0017	From pressure coefficients of standards and typical atmospheric changes during test
Combined type B uncertainty	0.0179	RSS of above components ($k = 1$)

$$u_A = \sqrt{\frac{\sum_{i=1}^{m}(n_i - 1)s_i^2}{\sum_{i=1}^{m}(n_i - 1)}} \tag{10.40}$$

where n_i is the number of measurements made on day i, and s_i the standard deviation of the m measurements on that day. Degrees of freedom for the Type A uncertainty is $\sum_{i=1}^{m}(n_i - 1)$.

For the data shown in Table 10.4, the Type A uncertainty ($k = 1$) is 0.00862 μΩ/Ω with 195 degrees of freedom.

10.8.2.3 Drift Uncertainty

When determining an uncertainty component to account for drift over the next calibration interval, the methodology described in Sect. 10.7 can be used. Historical data on the resistor since 1990 is plotted and a linear regression is performed (Fig. 10.15). For ease of calculation in this case study, the time-of-test uncertainty was assumed to be a constant $\sigma_y = 0.020$, (RSS of the Type A and Type B components found above) for all years. Complete analysis would include a varying uncertainty for each year to account for equipment changes, etc. and would require a weighted linear regression.

The regression results in a drift rate of 0.031 μΩ/Ω/year, with uncertainty $\sigma_B = 0.0011$ μΩ/Ω/year. With $N = 19$ historical data points and $\bar{t} = 732168$ (the average of all historical calibration dates), substituting the values into the prediction band formula (see Sect. 10.3.3), and simplifying gives the $k = 1$ uncertainty at any point:

$$u_{\text{pred}}(t) = \sigma_y \sqrt{1 + \frac{1}{N} + (t - \bar{t})^2 \left(\frac{\sigma_B}{\sigma_y}\right)^2}$$

Table 10.4 Type A uncertainty measurements

Day	Average deviation from nominal (μΩ/Ω)	Standard deviation	Number of measurements
1	0.197	0.0075	40
2	0.217	0.0062	40
3	0.198	0.0095	40
4	0.199	0.0101	40
5	0.196	0.0092	40

$$\sigma_c^2 = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} u_{\text{conf}}^2(t)dt = \sigma_B^2\left(\frac{t_1^2}{3} + \frac{t_1 t_2}{3} - t_1\bar{t} + \frac{t_2^2}{3} - t_2\bar{t} + \bar{t}^2\right) = 0.000195. \tag{10.44}$$

The drift uncertainty, u_R, is then given as the RSS combination of the two components:

$$u_R = \sqrt{\sigma_{\bar{b}}^2 + \sigma_c^2} = 0.0141\ \mu\Omega/\Omega. \tag{10.45}$$

The time-of-test uncertainty $\sigma_{\bar{y}}$, as listed in Sect. 10.8.1, is omitted from the drift uncertainty u_R and included in the next section. The degrees of freedom for a fitted line are $N-2$ where N is the number of measurement values ($y_i's$), and 2 is subtracted to account for the 2 parameters that were fit (slope and intercept). In this case, degrees of freedom are 17.

10.8.2.4 Expanded Uncertainty

The total combined uncertainty is the RSS combination of Type A and Type B time-of-test uncertainties and the drift uncertainty. The effective degrees of freedom are calculated via the W-S formula and come out large enough in this case to use $k = 2.0$ for a 95.45% confidence interval (Table 10.5).

Therefore, the reported resistance deviation from nominal is the mean of the 5 days of measurements, and the total expanded uncertainty is

$$0.201\ \mu\Omega/\Omega \pm 0.048\ \mu\Omega/\Omega (95.45\%, k = 2.0). \tag{10.46}$$

However, at the time of the next calibration, the resistor will be evaluated for OOT conditions based on the prediction band formula calculated on the expiration date, not the single interval uncertainty value of 0.048.

Table 10.5 Combined uncertainty components

Component	$k = 1$ Uncertainty (μΩ/Ω)	Degrees of freedom
Type B	0.0179	∞
Type A	0.00862	195
Drift allowance	0.0141	17
Combined (RSS)	0.024	2506

10.9 Calibration Interval Analysis

Interval analysis, determining the optimal length of a calibration interval, is another application of curve fitting in metrology. Many calibrations expire after 1 year. While convenient from a calendar perspective, this may not be the optimal interval from a risk perspective. If the calibration interval is set too long, a device is more likely to drift out of tolerance before the next calibration. If too short, there will be increased costs due to extra calibrations and equipment downtime.

Typically, interval adjustment is performed on devices that are tolerance tested to a specification limit such as a manufacturer's specification. The goal of interval analysis is to find the longest interval that results in the device remaining within its limits to some given probability during the interval. A good interval analysis requires a sufficient amount of historical data. Frequently, data from an individual device will not be sufficient, but when the data from all devices of the same model are combined, better results can be obtained.

Depending on what type of historical records are available for class of devices, there are three common methods for determining an interval. Sometimes, only the pass/fail status is recorded in a calibration database without keeping any information on exactly where a device measured within or outside the limits. In this case, if all the calibration intervals in the historical record were approximately the same length, the Interval Test method can be used. It is commonly referred to as "Method A3" due to its designation in NCSLI Recommended Practice 1 (RP-1) (NCSLI 2010). When the historical data recorded only pass/fail status but the intervals were of varying length, the Binomial Method (often referred to as "Method S2" from RP-1) can be used for better results. The Variables Method can be used if historical data contains the actual value measured in addition to pass/fail status, providing the most accurate results (Castrup 2005). The Variables Method and Method S2 both involve fitting a curve to the historical data. Methods A3 and S2 are fully described in RP-1, so the remainder of this section will focus on the Variables Method.

The first step in performing an interval analysis using the Variables Method is to convert the historical data from time of calibration (t) and measured value (y) into deltas Δt and Δy, the differences between subsequent calibrations. Note that if any adjustments are made during the calibration, Δy values should be calculated as the difference between as-found value of one calibration and as-left value of the previous calibration.

Once the historical data is in (Δt, Δy) pairs, fit a curve to the points. In theory, the curve could be any model, such as polynomial or exponential, but it is typically a straight line with intercept forced to zero. This assumes the device is adjusted back to the center of the specification limits after each calibration. Determining an optimal interval then becomes a matter of calculating the prediction band at the desired confidence level and solving for the time at which the prediction band exceeds the specification limits.

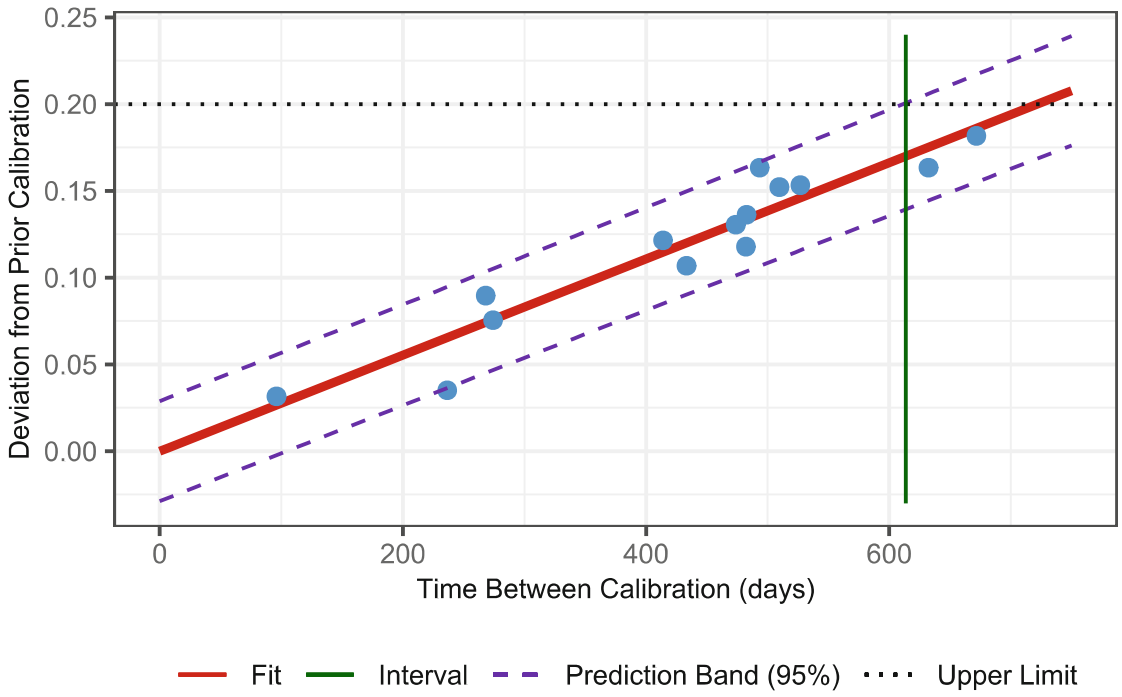


Fig. 10.16 Interval analysis for a voltage source

10.10 Summary

Fitting a curve to data is a common exercise in metrology, yet a proper evaluation of uncertainty in the fitted curve is often neglected. This chapter has provided general guidance on how uncertainties can be determined in both the fitted parameters and in the curve as a whole. The benefits of a careful assessment of fitted curve uncertainties can be great. For example, uncertainties involving equipment with temperature dependencies can be significantly reduced when the temperature dependency is fit to a curve. The temperature can then be measured during each use of the equipment, and the uncertainty at that temperature can be used rather than a worst-case uncertainty. Using curve fitting techniques, in combination with the risk analysis of Chap. 5, also provides a means of predicting the probability of future OOT conditions.

It should always be verified that the model being fit to the data (linear, polynomial, exponential, etc.) makes physical sense and represents the data without overfitting. Sample sizes should also be chosen to ensure an adequate number of degrees of freedom are available to estimate overall uncertainty. Additional points along the fitted curve not only increase total degrees of freedom, but they reduce the uncertainties in fitted parameters. For more recommendations on degrees of freedom and sample size, see Sect. 11.5.

10.11 Related Reading

While the GUM says very little about curve fitting, the BIPM has announced a document in preparation titled "Applications of the least-squares method" which, once released, promises to be a definitive resource on curve fitting as applied to metrology. ISO/TS 28037:2010 provides guidance on use of straight-line fits for calibration curves, and ISO/TS 28038:2018 extends that guidance to polynomial calibration curves. NCSLI's Recommended Practice 1 (RP-1) applies regression analysis to the determination of suitable calibration intervals for managing risk.

Most nonlinear curve fitting problems apply a numerical minimization technique to find the best fit parameters. The Gauss–Newton method is described in Hartley (1961), and the Levenberg–Marquardt algorithm described by Moré (1978). Both are implemented in modern computer code by Press et al. (2002).

Concerns over selecting the correct model (polynomial vs. exponential fit, appropriate order of polynomial, etc.) to prevent under- or overfitting, and validating the adequacy of the model itself are discussed by Harrell (2015). Izenman (2013) provides an in-depth discussion of how to extend the concepts introduced in this chapter to the multivariate case—when there are more dimensions than just x and y in the regression problem. Details of the Orthogonal Distance Regression algorithm are given in Boggs and Rogers (1990). Carroll and Ruppert (1996) cautions the user on use of ODR when the measurement equation introduces its own error. An analysis of least-squares fitting applied to coordinate measuring machines (CMMs) is given by Srinivasan (2012), which considers fitting measured CMM points to lines, planes, and other curved surfaces to evaluate the dimensions of the object under test. A treatment of the often-ignored effects of autocorrelation (correlation in time between uncertainties in all x values, for example) is provided in King and Giles (2018).

Regression problems can also be approached from a Bayesian perspective, accounting for a priori information. Klauenberg (2015) gives a tutorial on a Bayesian approach to regression, and Elster and Wübbeler (2016) provides an example and comparison to traditional least-squares regression. Markov-Chain Monte Carlo also applies a Bayesian analysis. For the statistics behind the algorithm as applied to regression problems, see Elster and Toman (2011). For an engineering application of MCMC to fitting X-ray diffraction spectra, see Iamsasri et al. (2017).

10.12 Exercises

1. A pressure gauge has been calibrated five times using a known 180 psi reference with results shown in Table 10.7.

   (a) Assuming the gauge is drifting in a straight line, find the best fit slope and intercept using Eq. (10.2).
   (b) Use Eq. (10.3) to determine the uncertainty in this slope and intercept assuming each measurement had a total ($k = 1$) uncertainty of 0.2 psi.

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